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Original article

The efficacy and safety of low-molecular-weight heparin use for cancer treatment: A meta-analysis

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ABSTRACT

Background: Low-molecular-weight heparin (LMWH) has an anti-tumour effect in-vitro and in animal models of malignancy; however, the evidence from clinical trials is controversial. Thus, we performed a meta-analysis from the results of randomised controlled trials (RCTs) to assess LMWH efficacy and safety in cancer patients who had no venous thromboembolism (VTE).

Methods: We searched the MEDLINE, EMBASE and CENTRAL (The Cochrane Central Register of Controlled Trails) databases covering all papers published up until April 2012. Two reviewers (D. H. Che and J. Y. Cao) extracted the data independently. The inclusion criteria used were patients with cancer who had no VTE and were treated with LMWH. The outcomes of interest included the 1-year mortality rate, VTE, bleeding and major bleeding complications. The results were presented as a relative risk (RR), and the STATA 11.0 package was used for comprehensive quantitative analysis.

Results: A total of 11 studies with 3835 cases and 3449 controls were included. The meta-analysis showed significant differences in the rates of bleeding with an RR: 1.32 [95% confidence interval (95% CI, 1.08–1.62)] and VTE with an RR: 0.53 (95% CI, 0.42–0.67) in cancer patients when LMWH was compared with placebo or no anticoagulant. There were no significant differences in the 1-year mortality rate with an RR: 0.97 (95% CI, 0.92–1.02) and major bleeding with an RR: 1.22 (95% CI, 0.87–1.71).

Conclusion: LMWH does not significantly reduce the 1-year mortality rate for cancer patients. Although LMWH can prevent VTE, we should consider the risk-effect ratio (in case of an increased bleeding event) when we use LMWH in the patients with cancer. Thus, further research is still needed to confirm these results.

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1. Introduction

Since Trousseau's initial observations in 1865, the relationship between thrombosis and cancer has been widely observed. As for thrombosis, venous thromboembolism (VTE) is an unwanted complication for cancer patients [1]. VTE in patients with cancer is strongly associated with reduced survival [2,3].

The pathogenesis of cancer-related VTE is complex, involving multiple interactions between malignant cells, endothelial cells and the coagulation cascade. Tumour cells can directly activate the clotting cascade through the release of procoagulant molecules (such as the tissue factor) and indirectly by the release of inflammatory cytokines [4–6]. Additionally, the breakdown of the vascular endothelium and activation of leukocytes and platelets leads to the formation of blood clots [4]. Moreover, other factors such as surgery, indwelling venous catheters and chemotherapy can further enhance the hypercoagulant process.

Because there is an intimate relationship between VTE and the development of cancer, anticoagulant therapies have demonstrated a beneficial effect on the cancer patients.

Low-molecular-weight heparin (LMWH) manufactured by the depolymerisation of standard heparin is capable of modifying tumour growth by several possible mechanisms. These include the inhibition of tumour growth factors, angiogenesis, generation of heparinase and thrombin, the reversal of multidrug resistance and other possible mechanisms [7,8].

Many studies have indicated that LMWH might improve the survival rates in certain populations of cancer patients [9,11,24,25]; however, other authors have reported conflicting results [10,12,16–19]. Because of this opposing evidence, it is uncertain whether the use of LMWH will benefit cancer patients. Therefore, we performed a meta-analysis from the results of randomised clinical trials (RCTs) to assess LMWH efficacy and safety in cancer patients who have no VTE.

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2. Materials and methods

2.1. Literature search strategy

We conducted a search of the MEDLINE, EMBASE and CENTRAL (The Cochrane Central Register of Controlled Trials) databases covering all papers published up until April 2012 with the following terms: "cancer or tumour or malignancy" and "LMWH or enoxaparin or dalteparin or reviparin or certoparin or tinzaparin or bemiparin or nadroparin or semuloparin" and "RCTs". The citation lists retrieved in the research were used to identify other papers with potential relevance.

2.2. Inclusion

Inclusion criteria were as follows: 1) cancer patients without a concurrent diagnosis of VTE; 2) RCTs; 3) treatments comparing LMWH with placebo or no intervention and 4) outcomes: VTE, bleeding, major bleeding and 1-year mortality rate.

2.3. Exclusion

Studies meeting the following criteria were excluded: 1) non-randomised studies; 2) non-cancer patients; 3) cancer patients with a concurrent diagnosis of VTE; 4) patients who had anticoagulation treatment including heparin and vitamin K antagonists such as warfarin and 5) duplicate publications, letters and review articles.

2.4. Data extraction and quality assessment

The data were manually extracted from each report independently generated by D. H. Che and J. Y. Cao, who were blinded from each other and used a standard data entry form. Any disagreements were resolved through discussion and in consultation with a third reviewer (Y. Yu). When the original data needed for the analysis could not be found in the published papers, the authors were asked to provide them. The following data were collected from each study: 1) the author's name; 2) the year of publication; 3) the number of patients included in the study; 4) the cancer characteristics; 5) the intervention (types, doses and lengths of LMWH treatment) and 6) the design and Jadad score.

The quality of the studies was appraised by the method previously described by Jadad et al., a validated quality assessment instrument [13,14]. This instrument evaluates the study quality based on the methods of randomisation, binding and description of study withdrawals. The possible scoring range was from 0 to 5 (score ≤ 2 = low quality and score ≥ 3 = high quality).

2.5. Statistical analysis

The cancer mortality rate, VTE, bleeding and major bleeding for the different treatment arms were used to calculate a separate pooled relative risk (RR) for each trial. These RRs were combined across studies, giving weight to the number of events in each of the two groups in each separate study. Statistical heterogeneity was measured by the Higgins I^2 statistic [15], which was defined as low if <25%, moderate if between 25% and 50% and high if >50%. A fixed-effects model was used when no heterogeneity existed among studies, otherwise the random-effects model was involved. Sensitivity analyses were performed to determine the stability of pooled results. Begger's funnel plots and tests were used to assess publication bias statistically ($p<0.05$ was considered representative of significant publication bias), while the STATA 11.0 software was used for meta-analysis.

3. Results

3.1. Description of studies

The systematic literature search yielded a total of 11 studies according to the selection criteria [9–11,16–23]. The flow diagram (Fig. 1) showed the process of selecting studies included in this meta-analysis. A total of 7284 patients with solid tumours received LMWH versus placebo or no intervention. The important details of the included studies are summarised in Table 1. The patients of all studies except one [18] had a different advanced solid malignancy. The concomitant treatment included chemotherapy, radiotherapy, hormone therapy or surgery. Six studies were placebo-controlled [10,11,18,19,22,23], and one placebo-controlled study was switched to a non-placebo-controlled protocol after it was identified as a major obstacle for adequate patient accrual [16]. Two studies [9,20] reported Jadad scores of 2 and others with more than 3.

3.2. Mortality

Eight studies [9–11,16–19,23] reported figures for the 1-year mortality rate. According to the fixed-effects model, LMWH did not significantly reduce the 1-year mortality rate with an RR: 0.97 [95% confidence interval (95% CI, 0.92–1.02)] (Fig. 2). In combination, these studies showed that 1450 of 3168 patients (46%) died of cancer in the LMWH group versus 1333 of 2781 patients (48%) in the control group. The heterogeneity was low ($I^2 = 20.3\%$).

3.3. VTE

There were 11 trials that assessed VTE [9–11,16–23]. A total of 106 of 3829 patients (2.8%) in the LMWH group versus 183 of 3436 patients (5.3%) in the control group experienced an episode of VTE. The pooled studies showed a significant reduced risk of VTE in the LMWH group (RR: 0.53; 95% CI, 0.42–0.67) (Fig. 3). The heterogeneity was low ($I^2 = 5.9\%$).

3.4. Bleeding

In 11 studies [9–11,16–23], bleeding occurred in 224 of 3815 (5.9%) patients in the LMWH group versus 144 of 3425 (4.2%) patients in the control group. LMWH was associated with a significant increase in bleeding (RR: 1.32; 95% CI, 1.08–1.62) (Fig. 4). The heterogeneity was moderate ($I^2 = 40.4\%$).

3.5. Major bleeding

Ten trials reported major bleeding [10,11,16–23], and from these studies, there was no apparent increase in major bleeding with LMWH use (RR: 1.22; 95% CI, 0.87–1.71) (Fig. 5). The pooled studies described major bleeding complications in 72 of 3773 patients (1.9%) in the LMWH group versus 57 of 3383 patients (1.7%) in the control group, and the heterogeneity was low ($I^2 = 9.2\%$).

3.6. Sensitivity analyses and publication bias

Sensitivity analyses using a fixed-effects model suggested that the exclusion of two low quality studies did not change the results of the primary analysis. The Begger's funnel plot and Begger's test were conducted to assess the publication bias. The funnel plot revealed a symmetrical appearance. Statistical results did not demonstrate publication bias ($p > 0.05$ for all) (Fig. 6).

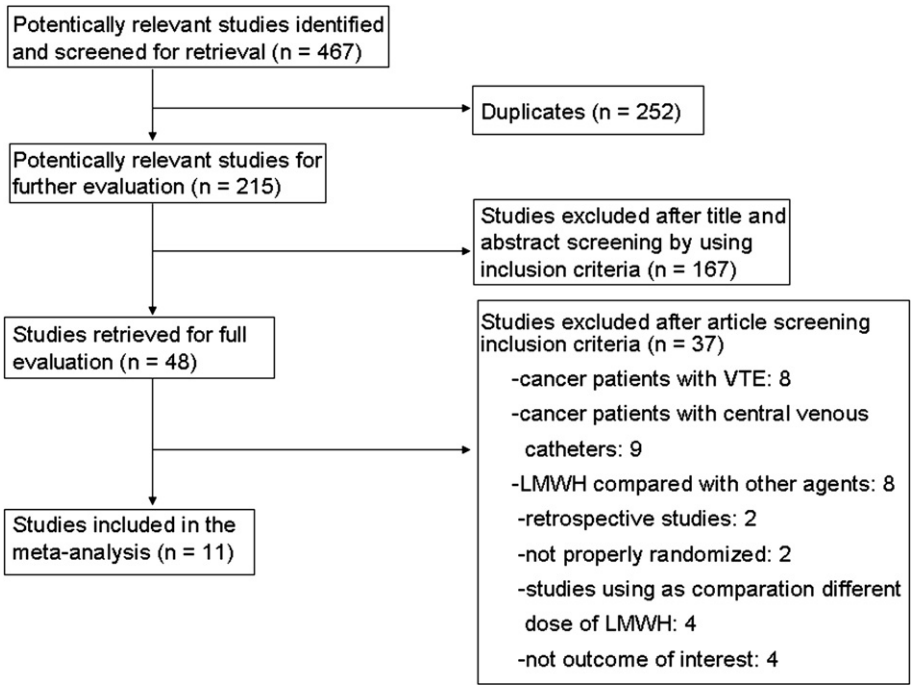

Fig. 1. Study flow of the systematic review. LMWH indicates low-molecular-weight heparin; VTE, venous thromboembolism.

4. Discussion

The meta-analysis included RCTs to evaluate the efficacy and safety of LMWH in the cancer patients who had no VTE. This systematic review revealed that LMWH did not significantly reduce the 1-year mortality rate in patients with cancer. A significant reduction in VTE was observed with the LMWH group. Not surprisingly, LMWH increased the risk of bleeding; however, major bleeding complications were not significantly increased.

Lazo-Langner et al. [24] focused their meta-analysis on RCTs that compared LMWH treated patients with a placebo group. These trials enrolled a total of 854 patients. At one year, the RR of death was 0.87 (95% CI, 0.77–0.99; $p = 0.04$). Another meta-analysis included 11 studies with cancer patients who had no VTE treated with LMWH

Table 1
Baseline characteristics of individual trials.

Study	No. of Patients	No. of controls	Inclusion criteria	Interventions	Lengths of LWMH	Design	Jadad score
Altinbas et al. 2004 [9]	84	42	Small cell lung cancer, previously untreated (all stages: 57% limited, 43% extensive)	Dalteparin 5000 U OD	18 weeks	Open-label randomised controlled trial	2
Kakkar et al. 2004 [10]	374	184	Advanced stage (III–IV) solid tumors	Dalteparin 5000 U OD	1 year or until death	Double-blind randomised placebo-controlled trial	5
Klerk et al. 2005 [11]	302	154	Advanced stage solid tumors (Mainly metastatic disease)	Nadroparin (3800–7600 U) BID for 2 weeks and OD for 4 weeks	6 weeks	Double-blind randomised placebo-controlled trial	5
Sideras et al. 2006 [16]	138	70	Advanced incurable solid malignancy	Dalteparin 5000 U OD	2 years or until death	Double-blind randomised placebo-controlled trial to open-label randomised controlled trial	4
Doormaal et al. 2011 [17]	503	259	Non-small-cell lung cancer (stage IIIB), hormone-refractory prostate cancer, locally advanced pancreatic cancer	Nadroparin (2 weeks at the therapeutic dose, and 4 weeks at half therapeutic dose)	6 weeks	Open-label randomised controlled trial	3
Perry et al. 2010 [18]	186	87	Malignant glioma	Dalteparin 5000 U OD	6 months	Double-blind randomised placebo-controlled trial	5
Agnelli et al. 2009 [19]	1150	381	Metastatic or locally advanced solid cancer	Nadroparin 3800 U OD	4 months	Double-blind randomised placebo-controlled trial	5
Riess et al. 2009 [20]	312	152	Advanced pancreatic cancer	Enoxaparin 1 mg/kg OD	6 months	Open-label randomised controlled trial	2
Maraveyas et al. 2011 [21]	123	63	Advanced pancreatic cancer	Dalteparin 200 U/kg OD for 4 weeks followed 150 U/kg for 8 weeks	12 weeks	Open-label randomised controlled trial	3
Haas et al. 2012 (TOPIC 1) [22]	353	179	Metastatic breast cancer	Certoparin 3000 U OD	6 months	Double-blind randomised placebo-controlled trial	4
Haas et al. 2012 (TOPIC 2) [22]	547	274	III/IV non-small-cell lung cancer	Certoparin 3000 U OD	6 months	Double-blind randomised placebo-controlled trial	4
Agnelli et al. 2012 [23]	3212	1640	Metastatic or locally advanced solid cancer (lung, pancreatic, stomach, clon, rectum, bladder, ovarian cancer)	Semuloparin 20 mg OD	3.5 months	Double-blind randomised placebo-controlled trial	4

U: units; OD: once daily; BID: twice daily; TOPIC: Thrombosis Prophylaxis in Oncologic Patients with Certoparin studies.

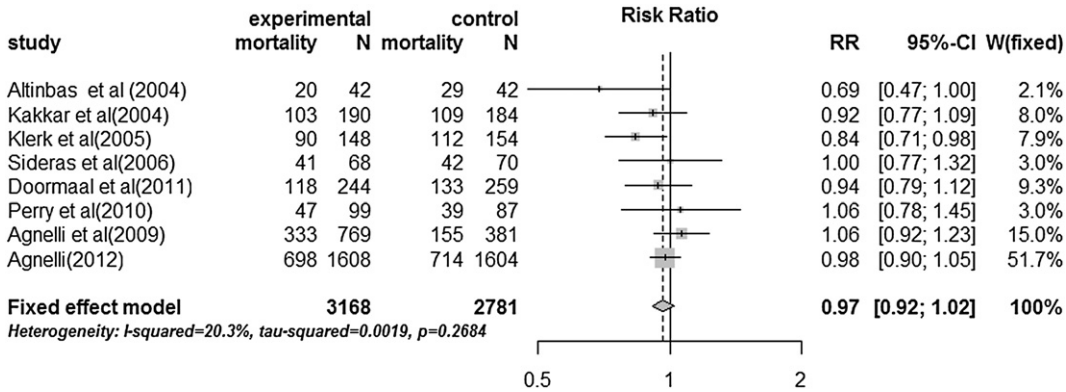

Fig. 2. Forest plot for the 1-year overall mortality rate in cancer patients randomised to LMWH versus placebo/no intervention.

[25]. The RR for 1-year mortality was 0.88 (95% CI, 0.79–0.98; $p = 0.015$) for LMWH.

In our systematic review, the RR for 1-year mortality rate is 0.97 (95% CI, 0.92–1.02). Although the outcome is not completely consistent with that found in the literature and has no significant difference, LMWH exhibits a favourable trend in the reduction of the 1-year mortality rate from the 95% CI of the RR.

What are the reasons that lead to such a small divergence? Several possible explanations may answer this question.

First, our meta-analysis included more patients and RCTs than had previously been published. Thus, some confounding factors such as the different histological types of cancer, stage of disease, life expectancy and different dose, duration and type of LMWH were not excluded. Second, the death of some patients included in the studies was possibly

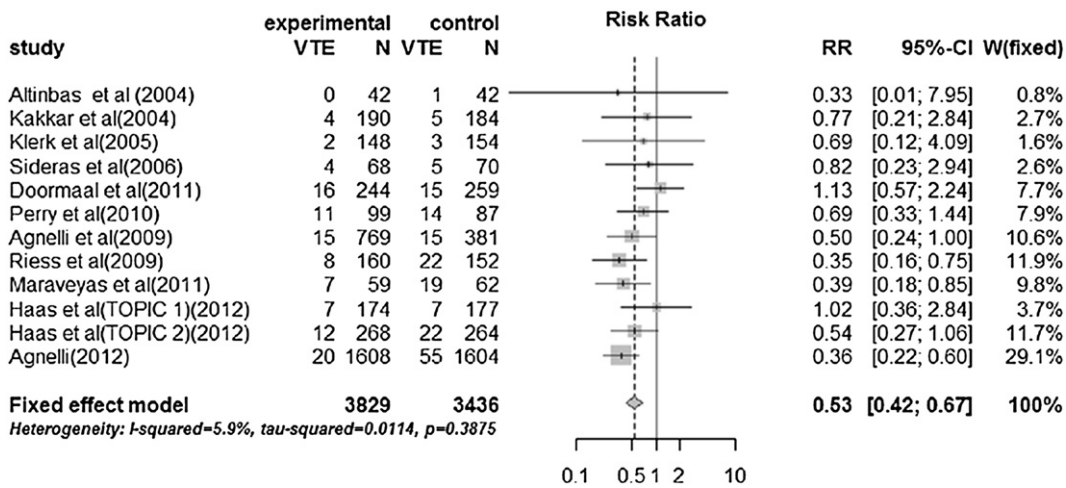

Fig. 3. Forest plot for VTE in cancer patients randomised to LMWH versus placebo/no intervention.

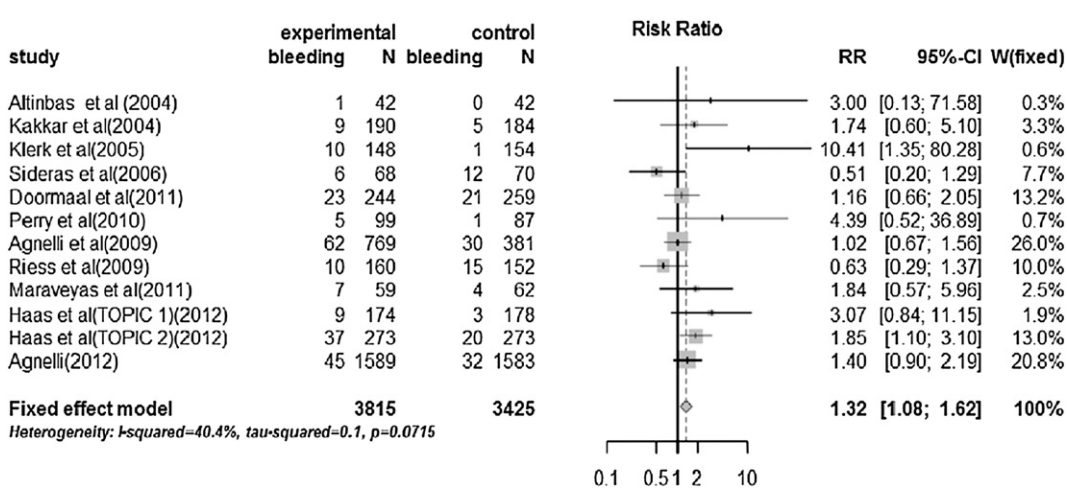

Fig. 4. Forest plot for bleeding in cancer patients randomised to LMWH versus placebo/no intervention.

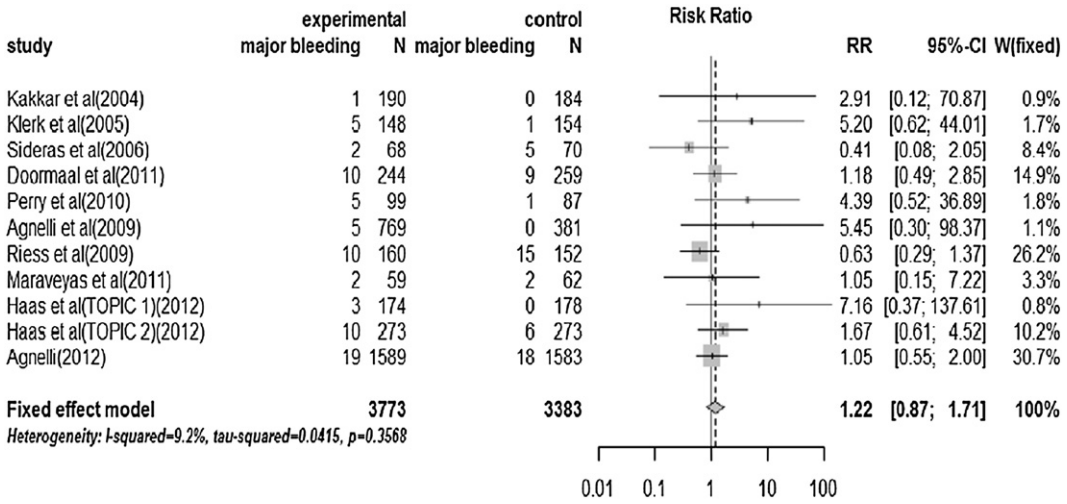

Fig. 5. Forest plot for major bleeding in cancer patients randomised to LMWH versus placebo/no intervention.

unrelated to VTE and could have been the result of many other causes. Third, many RCTs did not have a 1-year mortality rate as the primary end point. Finally, this systematic review included patients with metastatic or locally advanced disease, whereas most of the benefits from LMWH were observed in patients with less advanced disease.

Three important RCTs (CANTHANOX [26], CLOT [27] and LITE [28]) clearly demonstrated that long-term use of LMWH was efficient for the treatment of VTE in cancer patients. Although our review included more studies and patients (11 studies and 7284 cancer patients), our results were similar to those previously reported.

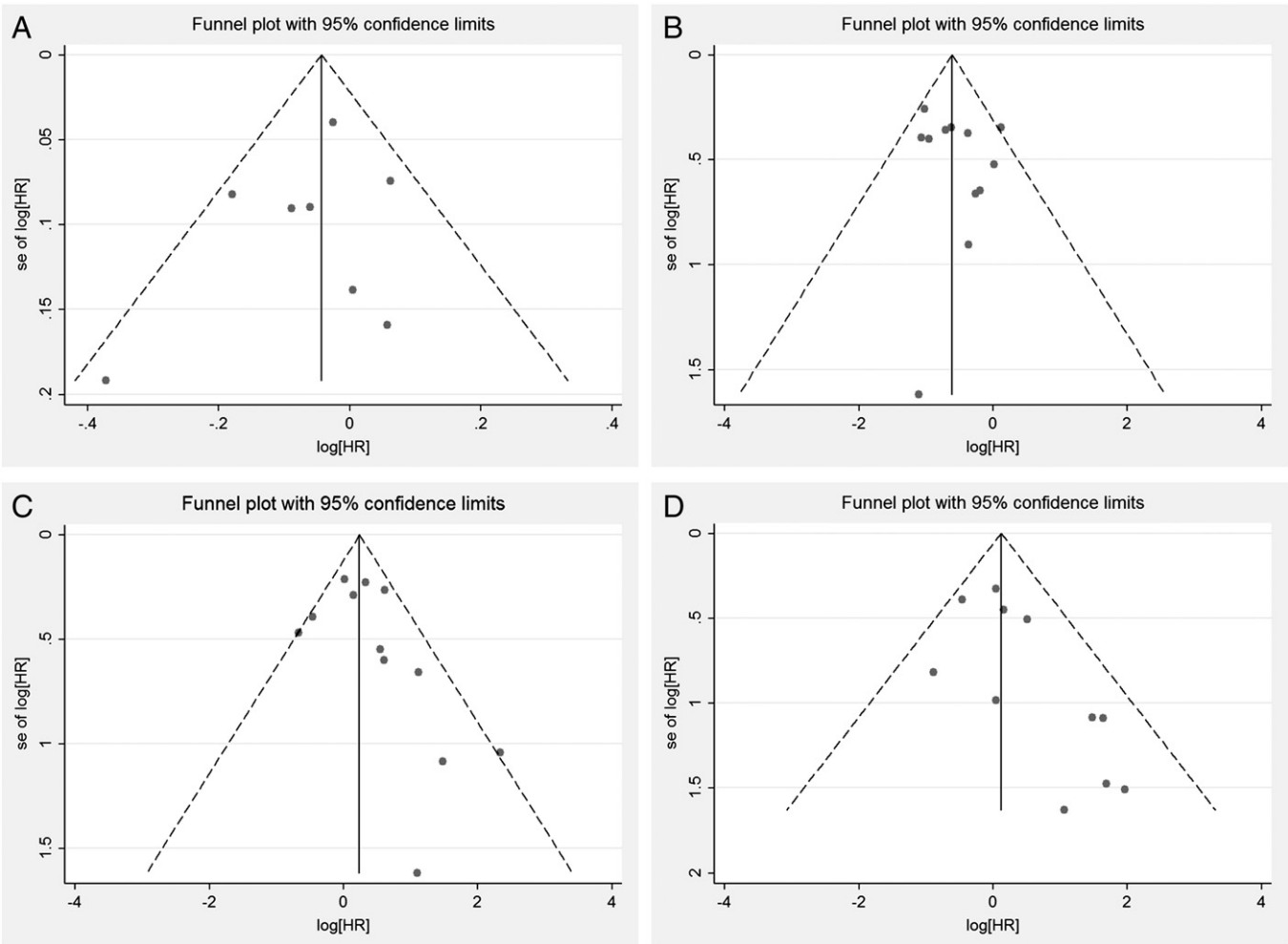

Fig. 6. Inverted funnel plots for trials comparing the effect on (A) the 1-year overall mortality rate, (B) VTE, (C) bleeding and (D) major bleeding of LMWH versus placebo/no intervention in patients with cancer.

However, LMWH has potential disadvantages, including increased bleeding, health care costs, potential patient discomfort from daily subcutaneous injections and the possibility of not surviving to see the overall benefit.

Who should receive LMWH? The most recent guidelines of the National Comprehensive Cancer Network emphasise the need for the assessment of VTE risk in ambulatory patients with cancer and the need for randomised studies of patients with a favourable risk-benefit ratio [29].

LMWH has been claimed to prolong survival in cancer patients [9,11,24,25]; however, the survival benefit is only observed in those patients with limited and non-metastatic cancer. Additionally, LMWH treatment benefits are specific to unique histological types of cancer such as small cell lung cancer or pancreatic cancer.

These results cannot be generalised for all cancer patients.

To better elucidate the benefits of LMWH, more RCTs are required. Future trials should focus on homogenous patient populations who have the same type of cancer, stage of disease and have been treated with standard anticancer therapies. Furthermore, the on-going studies have to address some issues about the optimal dose, the regimen and the duration of the treatment with LMWH.

In conclusion, this systematic review demonstrates that LMWH is effective in preventing VTE in the treatment of cancer patients, at the same time increases bleeding complications. There are no significant effects on major bleeding occurrences or the 1-year mortality rate. Thus, we should consider the risk-effect ratio when LMWH is used in cancer patients. Further studies are still needed to confirm these results.

Author contributions

Y. Yu: concept, design, critical revision and approval of the article. D. H. Che and J. Y. Cao: data collection, data analysis/interpretation, drafting the article and statistical analysis. Y. Yu, Y. C. Man and L. H. Shang: statistical analysis and drafting the article. All authors participated in this meta-analysis and have seen and approved the final version.

Conclusion

LMWH does not significantly reduce the 1-year mortality rate for cancer patients. Although LMWH can prevent VTE, we should consider the risk-effect ratio (in case of an increased bleeding event) when we use LMWH in the patients with cancer. Thus, further research is still needed to confirm these results.

Learning points

- Since Trousseau's initial observations in 1865, the relationship between thrombosis and cancer has been widely observed. As for thrombosis, venous thromboembolism (VTE) is an unwanted complication for cancer patients.
- Low-molecular-weight heparin (LMWH) reduces the incidence of VTE in cancer patients from 5.3% in the control group to 2.8% in the LMWH group.
- LMWH does not significantly reduce the 1-year mortality rate in patients with cancer, at the same time increases bleeding complications; however, the episodes of major bleeding are rare.
- Physicians should consider the risk-effect ratio when LMWH is used in the patients with cancer.

Conflict of interests

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, "The efficacy and safety of low-molecular-weight heparin use for cancer treatment: A meta-analysis."

Acknowledgements

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